# Lesson 3: Take Your Shot

GameMaker

VIDEO GAME CREATION

This week: spawning objects at runtime. Your job is to make your player shoot a projectile that flies up the screen and cleans itself up when it leaves - the first piece of your capstone's actual shooting mechanic.

**Open your own [YourName]Game project from Lesson 2 - don't start a new one. Everything this week builds on the player object you already have.**

*Tip: this lesson's Assets folder has a ready-to-use placeholder projectile image (bullet_yellow.png) and a new sound effect (sfx_shoot.wav) - you don't need to find or make your own art today.*

## Setup

1. Open your [YourName]Game project from Lesson 2.
2. Create a new sprite spr_bullet by importing bullet_yellow.png from this lesson's Assets folder.
3. Create a new object obj_bullet, assign spr_bullet to it.

## Level 1 - Minimal (MVP)

*Pressing the fire key spawns one projectile that flies upward and disappears when it leaves the screen. This is a complete finish for today - reaching this is enough to move on to next week.*

4. Add a Step event to obj_bullet with the code below - it moves itself upward and destroys itself once it's off-screen.
5. On obj_player (from Lesson 2), add to its existing Step event: a check for keyboard_check_pressed(vk_space) that calls instance_create_layer() to spawn an obj_bullet at the player's position.
6. Drag obj_player into your room if it isn't already there, then press F5 (Run) and press spacebar to test - a bullet should spawn at the player and fly upward, then vanish.

```
// obj_bullet's Step event:
y -= 8;
if (y < 0) instance_destroy();

// Add to obj_player's existing Step event:
if (keyboard_check_pressed(vk_space)) {
    instance_create_layer(x, y, "Instances", obj_bullet);
}
```

## Alternate Path: GML Visual (no typing)

*The same spawn/move/cleanup pattern works with drag-and-drop blocks instead of typed code. Pick whichever path feels right for you - both are graded the same.*

7. On obj_bullet's Step event, switch to GML Visual. From the Movement category, drag "Jump To Point," set X to 0 and Y to -8, and check Relative for BOTH X and Y - Jump To Point is the dedicated block for moving relative to the current position, so prefer it over the generic Assign Variable/Set Variable block, but it only moves relative to the current position on the axes you actually check Relative for. Leaving X's Relative box unchecked (even with X set to 0) snaps the bullet to an absolute X of 0 every step. Then from Common, drag an "If Variable" block comparing y < 0, with an "Instance - Destroy Instance" block inside it.
8. On obj_player's Step event, add an "If Key Pressed" block (Mouse and Keyboard category, Key set to Space - this is the single-press version, not the held-down "If Key Down" block from Lesson 2), with an "Instance - Create Instance" block inside it: set the Object dropdown to obj_bullet, set both its X and Y fields to 0, and check Relative for BOTH X and Y so the bullet spawns at the player's own current position instead of the room's absolute (0, 0).
9. Drag obj_player into your room, then press F5 (Run) and test with spacebar - same result as the code path.

*Level 2/3: "If Key Pressed" already only fires once per press, the same as keyboard_check_pressed() in the code path, so no extra cooldown logic is required for Level 1/2. Level 3's ideas are all reachable with more blocks from the Audio and Instance categories - same ideas as the code path, just dragged instead of typed.*

## Level 2 - Complete (Alpha/Beta)

*Firing repeatedly produces multiple independent projectiles on screen at once, each moving and disappearing correctly, and rapid-fire is capped so holding the key down doesn't flood the screen.*

- Test firing several times quickly - each press should produce its own independent obj_bullet instance, all moving and destroying themselves correctly. Since instance_create_layer() creates a fully separate instance each call, this already works with no extra code.
- Add a cooldown so holding spacebar doesn't fire every frame: keyboard_check_pressed() already only returns true on the single frame a key is first pressed (not every frame held, unlike keyboard_check()), so Level 1 is already immune to this specific bug - add a fire_cooldown = 0; variable in obj_player's Create event anyway if you want a slower, tunable fire rate (count it down by 1 each Step, and only allow firing when it's <= 0, resetting it to e.g. 12 - about 0.2 seconds at 60fps - on each shot).

## Level 3 - Refined (Ready to Publish) - pick any that interest you

- Sound: add sfx_shoot.wav as a sound resource (keep its default name, sfx_shoot), then call audio_play_sound(sfx_shoot, 1, false) right next to the instance_create_layer() call.
- Visual: give obj_bullet its own image_blend color and a brief image_xscale/image_yscale pulse that reads clearly as "a shot" rather than a copy of the player, then fades back to normal size and color - not a permanent recolor. image_xscale/image_yscale scale around the sprite's Origin point, so set spr_bullet's

Origin to Center in the Sprite Editor first (Sprite Editor > Origin) - scaling around any other origin (GameMaker's default is top-left) visibly shifts the bullet right and down as it grows, instead of scaling evenly in place.

- Structure: keep everything about how a bullet behaves - its own movement, its own off-screen cleanup - inside obj_bullet's own Step event, and keep obj_player's Step event limited to just spawning it (instance_create_layer + cooldown + sound). Don't reach into a bullet's variables from the player's code, or vice versa.

## If Something's Wrong

- Nothing happens on spacebar: confirm the check is keyboard_check_pressed (fires once), not keyboard_check (fires every frame the key is held) - using the wrong one either does nothing visible different, or floods the room with bullets.
- Bullet spawns but doesn't move or never disappears: confirm the movement code is in obj_bullet's Step event, not its Create event - Create only runs once, when the instance is first spawned.
- "instance_create_layer" errors about a missing layer: confirm the room actually has an instance layer named "Instances" (GameMaker's default layer name) - check the Room Editor's Layers panel and adjust the layer name argument to match if it's been renamed.
- GML Visual only: bullet always spawns in the room's bottom-left or top-left corner instead of at the player: this means one of the Relative checkboxes on "Jump To Point" or "Instance - Create Instance" was left unchecked - every X and Y field on both blocks needs Relative checked, not just Y.

## Want More?

Student Resource List > GameMaker section, or gamemaker.io/en/tutorials